# Project Asteria - Robotics 101 (Enhanced)

## Topics Covered

• Robot architecture: Sensors -> Controller -> Actuators -> Power.

• Sense -> Think -> Act loop.

• Sensors convert physical phenomena into electrical/digital signals.

• Ultrasonic sensors measure time-of-flight; distance = (speed of sound × time)/2.

• Wheel encoders measure rotation using counts, not distance directly.

• Odometry motivation: encoder counts help estimate robot motion.

• DC motors convert electrical energy -> magnetic energy -> mechanical rotation.

• Main parts: stator, rotor (armature), commutator, brushes.

• Commutator reverses current every half turn to maintain continuous rotation.

• Motor does not stop instantly when power is removed because of rotational inertia and stored kinetic energy; friction and air resistance gradually dissipate energy.

• Moment of inertia is the rotational analogue of mass. Larger moment of inertia stores more rotational kinetic energy and takes longer to stop.

• PWM (Pulse Width Modulation): motor receives full supply voltage in rapid ON/OFF pulses.

• Duty cycle controls average delivered power and therefore average torque/speed.

• Higher PWM frequency mainly makes torque and motion smoother; increasing duty cycle increases available average torque.

## Key Takeaways

Think from first principles. Separate instantaneous behavior (12 V pulses) from average behavior (effective power). Build intuition before memorizing formulas.